

Optical Optimization of Semitransparent Perovskite Devices

Overview

This project investigated ways to improve the optical performance of semitransparent perovskite devices for next-generation optoelectronics.

What I did

I fabricated perovskite micro-arrays using laser micro-processing to quantify visible-light transmittance. I also designed Ag/MoO₃/Ag transparent electrodes and studied how the layer design changed chromaticity.

Approach

The work combined device fabrication and measurement with theoretical optical modelling. The modelling was used to direct the experimental programme, and progress was presented in weekly group meetings.

PHOTOVOLTAICS

THIN-FILM OPTICS

EXPERIMENT